

Product Requirements Document

Fleet Maintenance Coordination System

1. Product Summary

The Fleet Maintenance Coordination System helps fleet operators monitor vehicle maintenance needs, coordinate repairs, assign work to internal or external service providers, and track maintenance history across a vehicle fleet.

The system receives vehicle health data from telematics devices, allows drivers to report issues, helps fleet managers schedule preventive and corrective maintenance, and coordinates work orders with repair vendors or internal maintenance teams.

The product is intended for organizations that operate commercial vehicle fleets, service fleets, delivery fleets, municipal fleets, or field-service vehicle networks.

2. Product Goals

The product should:

1. Improve visibility into vehicle health and maintenance status.
2. Reduce unplanned vehicle downtime.
3. Help fleet managers prioritize maintenance work.
4. Coordinate maintenance assignments between drivers, internal teams, and external vendors.
5. Track work order status from issue detection through completion.
6. Maintain a reliable vehicle maintenance history.
7. Support preventive maintenance scheduling based on mileage, time, usage, or diagnostic events.
8. Provide operational alerts when vehicles require urgent attention.

3. Non-Goals

The product will not:

1. Replace telematics hardware or vehicle onboard diagnostic systems.
2. Perform mechanical diagnosis beyond interpreting available telemetry and reported symptoms.
3. Manage parts inventory as a full warehouse management system.
4. Process vendor payments or accounting reconciliation in the initial release.
5. Provide route planning, dispatch optimization, or driver payroll management.
6. Replace vendor shop management systems.
7. Provide autonomous control of vehicles.

8. Replace driver safety training, inspection policy, or regulatory compliance ownership.

4. Target Users

User	Description
Fleet Manager	Oversees vehicle health, maintenance priorities, work orders, vendors, and fleet availability.
Maintenance Coordinator	Schedules repairs, assigns work orders, communicates with vendors, and tracks completion.
Driver	Reports vehicle issues, submits inspection results, views maintenance instructions, and confirms vehicle availability.
Internal Technician	Performs maintenance work and updates work order status.
Repair Vendor	Accepts assigned work orders and provides repair updates.
Operations Manager	Reviews fleet availability, downtime, overdue maintenance, and maintenance impact on operations.
System Administrator	Configures vehicles, users, vendors, maintenance rules, notification rules, and integrations.

5. External Systems and Devices

External System or Device	Description
Telematics Device	Sends mileage, engine hours, location, fault codes, vehicle status, and usage telemetry.
Vehicle Diagnostic System	Provides diagnostic trouble codes and vehicle health events.
Vendor Management System	Optional system used by repair vendors to receive and update work orders.
Notification Provider	Sends maintenance alerts, work order updates, reminders, and escalations.
Identity Provider	Optional authentication source for fleet staff, technicians, and vendors.
Parts or Procurement System	Optional future integration for parts availability, purchase requests, or repair cost tracking.

External System or Device	Description
Accounting System	Optional future integration for vendor invoice or maintenance cost reconciliation.

6. Primary User Problems

Fleet operators often discover vehicle problems too late, after a breakdown or missed service interval. Maintenance work may be coordinated through phone calls, emails, spreadsheets, and vendor portals, making it difficult to track status, ownership, and fleet availability.

Drivers need a simple way to report vehicle issues. Fleet managers need a consolidated view of vehicle health, upcoming service needs, repair status, and historical maintenance records. Maintenance coordinators need structured work order workflows that support both internal technicians and external repair vendors.

7. Product Capabilities

7.1 Vehicle Registry

The system should maintain a registry of fleet vehicles.

Vehicle records may include:

1. Vehicle ID.
2. VIN.
3. License plate.
4. Vehicle type.
5. Make, model, and year.
6. Depot or region.
7. Assigned driver or operating group.
8. Current availability status.
9. Current odometer or mileage.
10. Current engine hours where applicable.
11. Telematics device association.
12. Maintenance policy group.
13. Active, inactive, retired, or sold status.

7.2 Vehicle Health Monitoring

The system should collect and display vehicle health signals from telematics and diagnostic sources.

Vehicle health signals may include:

1. Mileage.
2. Engine hours.

3. Diagnostic trouble codes.
4. Battery voltage.
5. Fuel or energy level.
6. Engine temperature.
7. Brake system warnings.
8. Tire pressure alerts.
9. GPS location.
10. Vehicle operating status.
11. Last telemetry timestamp.

The system should distinguish between current, stale, missing, and invalid telemetry.

7.3 Maintenance Alerting

The system should generate alerts when vehicles require maintenance attention.

Alerts may be triggered by:

1. Diagnostic trouble codes.
2. Preventive maintenance intervals.
3. Driver-reported issues.
4. Failed inspection checklist items.
5. Overdue service.
6. High-severity vehicle health events.
7. Repeated warning patterns.
8. Vendor or technician delay.
9. Vehicle unavailable beyond expected repair window.

Each alert should include severity, affected vehicle, likely maintenance category, supporting evidence, and recommended next action.

7.4 Preventive Maintenance Scheduling

The system should support preventive maintenance schedules based on:

1. Mileage.
2. Engine hours.
3. Calendar time.
4. Vehicle class.
5. Manufacturer recommendation.
6. Fleet policy.
7. Prior maintenance completion date.
8. Usage pattern.

Fleet managers should be able to configure maintenance templates for different vehicle types.

The system should identify upcoming, due, and overdue preventive maintenance items.

7.5 Driver Issue Reporting

Drivers should be able to:

1. Report vehicle issues from a mobile-friendly experience.
2. Select the affected vehicle.
3. Choose an issue category.
4. Enter a description.
5. Attach photos where supported.
6. Mark whether the vehicle appears safe to operate.
7. Submit inspection checklist results.
8. View status of issues they reported.

The system should convert driver-submitted issues into maintenance alerts or work order candidates.

7.6 Work Order Management

The system should support the full maintenance work order lifecycle.

A work order may include:

1. Work order ID.
2. Vehicle.
3. Reported issue or preventive maintenance reason.
4. Priority.
5. Assigned technician or vendor.
6. Target completion date.
7. Maintenance tasks.
8. Parts or service notes.
9. Labor notes.
10. Status.
11. Attachments.
12. Completion summary.
13. Vehicle availability impact.
14. Related diagnostic events or driver reports.

Work order statuses should include at minimum:

1. Draft.

2. Open.
3. Assigned.
4. Accepted.
5. In Progress.
6. Waiting on Parts.
7. Waiting on Approval.
8. Completed.
9. Canceled.
10. Rejected.

7.7 Assignment to Internal Technicians or External Vendors

Maintenance coordinators should be able to assign work orders to either internal technicians or external repair vendors.

The system should:

1. Track assignee type.
2. Notify assignees of new work.
3. Allow assignees to accept or reject work.
4. Track status updates.
5. Record estimated completion time.
6. Escalate overdue or stalled work.
7. Keep fleet managers informed of vehicle availability impacts.

External vendor integration may be manual through a vendor portal in the initial release, with system-to-system vendor integration in later releases.

7.8 Vehicle Availability Tracking

The system should track whether a vehicle is:

1. Available.
2. In service.
3. Assigned for maintenance.
4. Out of service.
5. Awaiting inspection.
6. Awaiting vendor update.
7. Waiting on parts.
8. Ready for pickup.
9. Retired.

Fleet managers should be able to view maintenance impact on fleet availability.

7.9 Maintenance History

The system should maintain a historical record for each vehicle, including:

1. Driver-reported issues.
2. Diagnostic events.
3. Preventive maintenance schedules.
4. Work orders.
5. Vendor assignments.
6. Technician notes.
7. Completion summaries.
8. Attachments.
9. Maintenance cost estimates where available.
10. Vehicle downtime intervals.
11. Vehicle availability status changes.

Maintenance history should support compliance reporting, resale evaluation, reliability analysis, and operational review.

7.10 Notifications and Escalations

The system should notify relevant users when:

1. A high-severity maintenance alert is created.
2. Preventive maintenance is due soon.
3. A work order is assigned.
4. A vendor or technician accepts or rejects work.
5. A work order status changes.
6. A vehicle is marked out of service.
7. A vehicle is ready to return to service.
8. A work order is overdue.
9. A driver-submitted safety issue requires immediate review.
10. A vehicle has stale or missing telemetry.

Notifications should be configurable by role, severity, vehicle group, location, and event type.

7.11 Reporting and Analytics

The system should provide reports for:

1. Fleet availability.
2. Preventive maintenance compliance.
3. Overdue maintenance.
4. Vehicle downtime.

5. Work order aging.
6. Vendor response time.
7. Maintenance frequency by vehicle.
8. Repeated fault patterns.
9. Driver-reported issue trends.
10. Maintenance cost estimates where available.
11. Vehicle availability by depot or region.
12. Technician or vendor workload.

Reports should support filtering by vehicle group, region, depot, vendor, vehicle type, date range, and maintenance category.

8. Representative User Journeys

8.1 Driver Reports Vehicle Issue

1. Driver opens the driver maintenance form.
2. Driver selects the assigned vehicle.
3. Driver selects issue category and enters a description.
4. Driver marks the vehicle as safe or unsafe to operate.
5. Driver submits the issue.
6. System creates a maintenance alert.
7. Maintenance Coordinator reviews the alert.
8. Maintenance Coordinator creates or links a work order.
9. Driver receives status updates.

8.2 Telematics Fault Creates Maintenance Alert

1. Telematics device sends a diagnostic trouble code.
2. System receives and validates the telemetry event.
3. System classifies the event severity.
4. System creates a maintenance alert.
5. Fleet Manager receives notification for high-severity events.
6. Maintenance Coordinator creates a work order.
7. Vehicle status is updated if the fault affects availability.

8.3 Schedule Preventive Maintenance

1. System evaluates vehicle mileage and service interval rules.
2. System identifies a vehicle due for preventive maintenance.
3. System creates a due-soon maintenance alert.
4. Maintenance Coordinator reviews available dates.
5. Maintenance Coordinator creates a preventive maintenance work order.
6. Work order is assigned to an internal technician or vendor.
7. Vehicle status is updated during service.
8. Completed maintenance resets the relevant service interval.

8.4 Assign Work Order to External Vendor

1. Maintenance Coordinator creates a work order.
2. Maintenance Coordinator selects an approved vendor.
3. System sends work order assignment to the vendor.
4. Vendor accepts or rejects the assignment.
5. Vendor updates work status.
6. Vendor marks work complete and adds service notes.
7. Maintenance Coordinator reviews completion.
8. Vehicle is returned to available status.

8.5 Fleet Manager Reviews Availability Impact

1. Fleet Manager opens the fleet availability dashboard.
2. System shows available, in-service, and out-of-service vehicles.
3. Fleet Manager filters by depot or vehicle class.
4. Fleet Manager identifies vehicles delayed in maintenance.
5. Fleet Manager reviews work order age and vendor status.
6. Fleet Manager escalates overdue work or adjusts operational planning.

9. Functional Requirements

ID	Requirement
FR-001	The system shall maintain a registry of fleet vehicles.
FR-002	The system shall collect telemetry from telematics devices where configured.
FR-003	The system shall process vehicle diagnostic events.
FR-004	The system shall identify stale, missing, or invalid vehicle telemetry.
FR-005	The system shall allow drivers to report vehicle issues.
FR-006	The system shall allow drivers to mark whether a vehicle appears safe to operate.
FR-007	The system shall generate maintenance alerts from telemetry, diagnostic events, preventive schedules, and driver reports.
FR-008	The system shall support configurable preventive maintenance schedules.
FR-009	The system shall identify upcoming, due, and overdue preventive maintenance.
FR-010	The system shall allow maintenance

ID	Requirement
	coordinators to create work orders.
FR-011	The system shall allow work orders to be assigned to internal technicians or external vendors.
FR-012	The system shall notify assignees of new work orders.
FR-013	The system shall allow assignees to accept, reject, or update work orders.
FR-014	The system shall track work order lifecycle status.
FR-015	The system shall update vehicle availability status based on maintenance state.
FR-016	The system shall maintain maintenance history for each vehicle.
FR-017	The system shall support attachments for issue reports and work orders.
FR-018	The system shall notify relevant users about high-severity alerts, due maintenance, work assignment, status changes, and overdue work.
FR-019	The system shall provide fleet availability reporting.
FR-020	The system shall provide preventive maintenance compliance reporting.
FR-021	The system shall provide work order aging and vendor performance reporting.
FR-022	The system shall maintain an audit trail of maintenance alerts, work order changes, assignments, status updates, and vehicle availability changes.
FR-023	The system shall support role-based access for drivers, fleet managers, maintenance coordinators, technicians, vendors, and administrators.
FR-024	The system shall expose failed integrations and stale telemetry to authorized operational users.

10. Non-Functional Requirements

Category	Requirement
Reliability	The system shall tolerate temporary telematics, vendor, or notification integration failures without losing maintenance state.
Availability	Fleet managers, coordinators, technicians, vendors, and drivers should be able to access maintenance workflows during fleet operating hours.
Data Freshness	Vehicle health views shall expose the timestamp and freshness status of telemetry.
Auditability	The system shall record who or what created, assigned, updated, completed, or canceled maintenance work.
Security	User access shall be authenticated and role-based. Vendor access shall be limited to assigned work orders.
Privacy	Driver information shall be visible only where needed for maintenance coordination and operational accountability.
Scalability	The system should support growth from a small fleet to a multi-region fleet with many vehicles and vendors.
Performance	Fleet dashboards and work order views should respond quickly under normal operating load.
Resilience	Telemetry, notification, and vendor workflows should support retry or recovery after temporary failures.
Observability	Operational staff shall be able to identify failed integrations, delayed work orders, alert backlogs, and telemetry gaps.
Configurability	Maintenance rules, alert thresholds, vehicle groups, vendor assignments, and notification policies should be configurable.
Data Retention	Maintenance history, diagnostic events, attachments, and audit logs shall be

Category	Requirement
	retained according to fleet policy and regulatory requirements.
Mobile Usability	Driver and technician workflows should be usable from mobile devices.

11. Data Requirements

The system should manage the following data:

1. Vehicle registry.
2. Vehicle groups, depots, regions, and assignments.
3. Driver references.
4. Telematics device registry.
5. Diagnostic event records.
6. Vehicle telemetry readings.
7. Preventive maintenance rules.
8. Maintenance alerts.
9. Driver issue reports.
10. Inspection checklist results.
11. Work orders.
12. Work order tasks.
13. Technician and vendor assignments.
14. Vendor references.
15. Vehicle availability status.
16. Attachments and photos.
17. Notification records.
18. Maintenance history.
19. Audit events.
20. Reporting aggregates.
21. User roles and permissions.

Vehicle maintenance history should preserve enough detail to explain when maintenance was needed, who acted, what work was performed, and when the vehicle returned to service.

12. Reporting and Operational Visibility

Fleet users should be able to view:

1. Fleet availability by depot, region, vehicle type, or group.
2. Vehicles currently out of service.
3. Vehicles due or overdue for preventive maintenance.

4. High-severity maintenance alerts.
5. Driver-reported issues awaiting review.
6. Open work orders by status.
7. Work orders assigned to each vendor or technician.
8. Work orders past target completion date.
9. Vendor response and completion performance.
10. Vehicle downtime trends.
11. Repeat issue patterns.
12. Telemetry freshness across active vehicles.

13. Success Metrics

Metric	Target Direction
Unplanned vehicle downtime	Decrease
Preventive maintenance completed on time	Increase
Vehicles with overdue maintenance	Decrease
Mean time from issue detection to work order creation	Decrease
Mean time from work order assignment to acceptance	Decrease
Mean time to return vehicle to service	Decrease
Driver-reported issues with status visibility	Increase
High-severity alerts acknowledged within target window	Increase
Vendor response time	Decrease
Repeat maintenance issues per vehicle	Decrease
Fleet availability rate	Increase
Telemetry freshness for active vehicles	Increase

14. Risks and Open Questions

Area	Question or Risk
Telematics integration	Which telematics providers and data formats must be supported first?
Diagnostic interpretation	How should diagnostic trouble codes be mapped to maintenance severity and action?
Vehicle safety	Who has authority to mark a vehicle unsafe or return it to service?
Vendor workflow	Will vendors use a portal, email notifications, or system-to-system integration?
Preventive maintenance rules	Are rules based on manufacturer

Area	Question or Risk
	schedules, fleet policy, usage pattern, or all of these?
Driver identity	Are drivers assigned to vehicles permanently, by route, or per shift?
Offline usage	Do drivers or technicians need offline access in low-connectivity environments?
Attachments	What file types and retention rules apply to photos, inspection forms, and repair documents?
Compliance	Are there regulatory inspection or maintenance record requirements for the fleet type?
Cost tracking	Should actual repair costs be captured in the initial release or deferred?
Vendor performance	Which vendor SLA metrics matter most?
Data retention	How long must diagnostic events, work orders, attachments, and audit history be retained?

15. Release Scope

MVP

The MVP should include:

1. Vehicle registry.
2. Driver issue reporting.
3. Basic telematics ingestion for mileage and diagnostic events.
4. Maintenance alert creation.
5. Preventive maintenance schedule configuration.
6. Work order creation and assignment.
7. Internal technician or manual vendor assignment.
8. Work order status tracking.
9. Vehicle availability status.
10. Email notifications for high-priority alerts and work assignments.
11. Basic maintenance history.
12. Fleet availability dashboard.
13. Preventive maintenance compliance report.
14. Audit logging for alerts, work orders, assignments, and availability changes.
15. Basic role-based access.

Post-MVP

Future releases may include:

1. Vendor portal.
2. Vendor system integration.
3. Mobile technician app.
4. Offline driver and technician workflows.
5. Advanced diagnostic code interpretation.
6. Parts inventory integration.
7. Cost tracking and budget reporting.
8. Predictive maintenance modeling.
9. Automated vendor selection.
10. SLA-based escalation.
11. Integration with procurement or accounting systems.
12. Advanced compliance reporting.
13. Multi-region fleet benchmarking.
14. Automated maintenance window recommendations.
15. Configurable inspection checklist builder.